

Kidney Disease Prediction Using Machine Learning:
A Comprehensive Data Analysis

Sonya Odeniyi

DePaul University

DREAM Project

12/4/2024

1.0 Introduction

Chronic Kidney Disease (CKD) has emerged as a significant health concern in today's world, affecting about 10% of the world's population, and this population is only projected to rise due to advances in life expectancy and a rise in the prevalence of critical ailments like Diabetes and Hypertension (Pakaya et al., 2021). Chronic Kidney Disease may progress to End-Stage Renal Disease (ESRD), which is associated with high mortality, high morbidity, and costly treatment modalities including dialysis or kidney transplantation (Youmans et al., 2023). Kidney transplantation (KT) continues to be the best treatment for ESRD due to better quality of life and survival rates compared to dialysis (Sharif, 2022). However, transplantation remains limited by donor availability, patient eligibility, and the ability to accurately predict successful transplant outcomes.

Given the growing number of patients needing a kidney transplant, it is crucial to establish the factors that influence both patient eligibility for the transplant and post-transplant survival. Predictors affecting the appropriateness of the kidney transplant include age, gender, ethnicity, complications, serum creatinine level, glomerular filtration rate, and donor organs available (Pan, 2023). Additionally, factors that influence post-transplant survival, such as graft survival rates and patient mortality, are equally important in ensuring long-term success. Machine learning (ML) has become one of the most effective tools for nephrology today, as it provides new algorithms for predicting transplant suitability, graft survival rates, and possible complications after transplant (Mulugeta et al., 2024). Kherab et al., (2024) employed Random Forests, Support Vector Machines, and logistic regression models to predict the probability of becoming a candidate for KT and to calculate the mortality rate after transplantations.

These models can aid clinicians in early patient risk stratification and lead to better decision-making of transplant appropriateness.

The objective of this study is to employ various machine learning techniques such as Random Forest, Support Vector Machine, and logistic regression model to a large dataset of clinical and demographic attributes of kidney transplant patients. We aim to assess the probability of patients achieving transplant status, while also considering factors that influence post-transplant survival, such as graft survival and patient mortality. By comparing the performance of these models, we hope to identify the most significant factors influencing both transplant eligibility and post-transplant outcomes, thereby improving patient management and long-term outcomes for individuals with ESRD.

2.0 Literature review

In the most recent years, machine learning (ML) methods have attracted much interest in the field of nephrology primarily in application to prognosis of kidney disease progression and transplant outcomes (Burlacu et al., 2020). Several studies have demonstrated the applicability of ML models in predicting essential clinical indicators such as initiation of dialysis, glomerular filtration rate (GFR), and the likelihood of kidney transplant eligibility. For instance, Random Forests have been employed in predicting dialysis beginning depending on clinical markers, including serum creatinine, GFR, and comorbid conditions (Godoi & Valan, 2024). By using these models, it has become evident that elements like age, blood pressure; and proteinuria among others are factors that predict the progress of kidney diseases as well as the eligibility for receiving a kidney transplant.

Similarly, logistic regression has been used in several studies as a useful approach to estimate the chances of being a candidate for kidney transplantation, using demographic and disease characteristics like age, race, diseases like diabetes, hypertension, and kidney functions (Chen et al., 2022). These models have offered understanding of risk factors concerning eligibility for transplant and have been applied to establishment of scoring system to evaluate patients for kidney transplantation. However, a major drawback of logistic regression features an interpretability of the model along with inability to handle nonlinear relationships and interactions characteristic to much clinical data as well, which is why more complex ML techniques like Random Forests and Support Vector Machines (SVM) appear to be more attractive.

A growing body of literature has also focused on post-transplant outcomes, particularly survival rates and graft failure predictions. Random Forest, support vector machine, and neural network have been applied to predict post-transplant mortality and graft survival where factors such as recipient age, donor details, immunosuppressant treatment, and laboratory values were incorporated (Gotileb et al., 2022). These studies exhibit the capacity of the ML models for identifying the complex interactions between variables while offering better predictive capability than traditional statistical models that are typically restrained by linearity and homoscedasticity assumptions.

However, only a handful of these studies have utilized large national level kidney transplant graft databases that contain data on both the pre-transplant eligibility and post-transplant graft survival in the same dataset. Many of the current investigations address either pre-transplant or post-transplant data, which hampers comprehension

of the patient experience. Furthermore, frequent tasks in the context of machine learning for predicting transplant eligibility fail to present a comparison of different model types, which can hamper the understanding of the pros and cons of various models. Following the prior studies of this algorithm, this paper utilizes Random Forest, Support Vector Machine, and logistic regression models for analysis on a big kidney transplant dataset in an effort to infer not only pre-transplant suitability but also post-transplant survival. We are interested in the set of attributes that can affect transplant success and analyze the performance of the chosen machine learning models.

Hence, our study adds to the body of knowledge by providing a comparison of such models comprehensively, and by identifying and prioritizing the key predictors of eligibility and survival after transplant along with avoiding many of the shortcomings of previous studies that exclusively relied on post-transplant data.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is an essential part of the process that aims to provide insight into the dataset and determine its features, relationships between variables, and possible shortcomings, including incorrect values, missing data, or skewed distribution. By analyzing the dataset in detail, we can define variables requiring additional preprocessing, select features for use in prebuilt machine learning models, and identify critical factors that influence the model creation process. This section briefly describes the EDA given for various variables in the given data set and presents detailed illustrations for such conclusions.

3.1 Summary Statistics

As a preliminary step to visualization, descriptive statistics were obtained for important variables including GFR, CREAT_CLEAR, INIT_AGE, and WGT_KG_TCR.

Descriptive statistics include measures of central tendency (mean, median) and variability (standard deviation, range). They serve as a basic premise for understanding visual patterns and provide the framework for decision-making about data transformation.

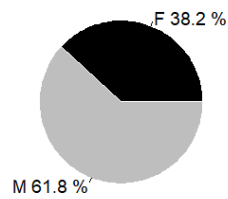
For example, the summary statistics of GFR showed that most patients had low kidney function while a vast majority of them were on the tail end of distribution. These may be as a result of medical anomalies or else contain data that is inaccurate. In the same fashion AGE and WEIGHT variables, had moderate to high skewness and thus may also need to be transformed before the modeling process.

3.2 Visualization

Data visualization is an essential step in the EDA process helping to gain an initial insight on data distribution. Data was analysed using different plots such as histogram plot, scatter plot, bar plot and density plot. From these visualizations, important findings were discovered and guided the next phase of data preprocessing.

1. Pie chart of Gender Distribution

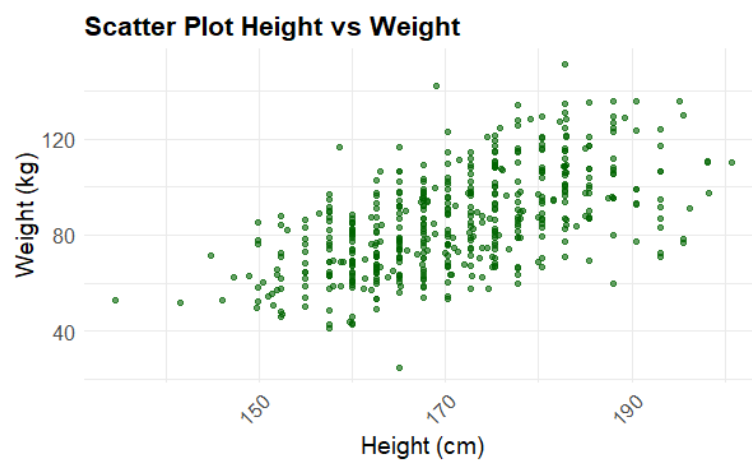
A pie chart was created to compare gender numbers. This visualization allowed comparing patients' gender distribution and can be considered essential for identifying the gender-specific characteristics of the disease progression.

Gender Distribution

Visual Insight: The pie chart showed that 38.2 % of the population were male and 61.8% were female

2. Height & Weight (Bar Graph)

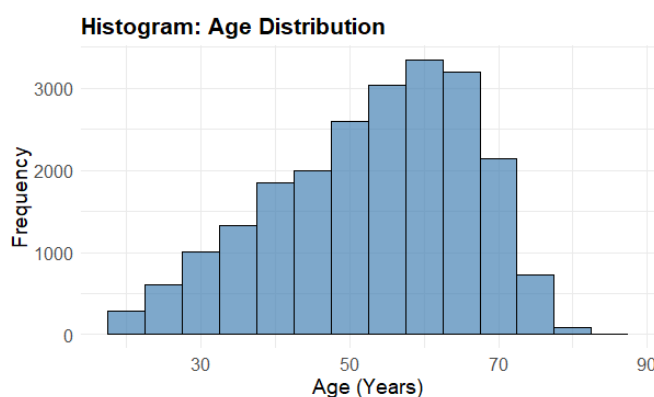
To examine the possible correlation between height and weight, we used a scatter plot. This plot showed a positive relationship which is normally the case in most populations.



Visual Insight: The scatter plot in case of height and weight positively correlated with each other and confirmed the routine physiological position between the height and weight of the person.

3. Age Distribution (Histogram)

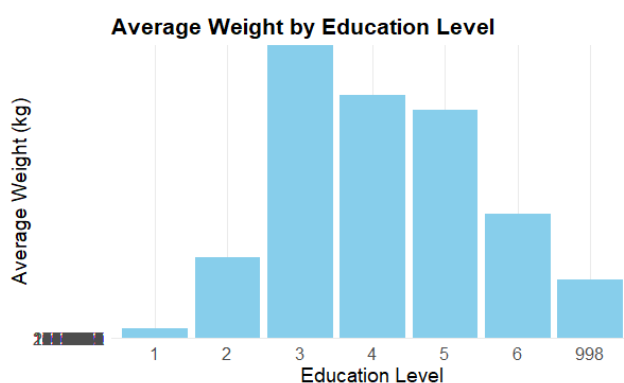
The next EDA plot was to make a histogram in order to determine the age of the patients in the provided dataset. The histogram provided a general picture where the majority of the patients' age was between forty and sixty years with a few of the patients being below forty or above sixty years. This age distribution is characteristic for chronic diseases such as kidney disease that tend to grow in frequency with age.



Visual Insight: The age distribution histogram confirmed that the disease is common among mid-elderly individuals. This realization brings out the significance of age in the analysis of the advancement and management of kidney ailments.

4. The Education Level by Region is presented in the Bar Plot below.

To analyze the differences of education level by regions a bar plot was developed. This plot underlined any regional differences in literacy levels and could also be correlated to regional health figures and access to healthcare.



Visual Insight: The bar plot again showed that there were differences in education in different regions which could be attributed to socio-economic determinants of health. This particular graphical presentation emphasizes the role of regionalism in kidney disease progression.

4.0 Methodology

4.1 Data Preprocessing

Data pre-processing is important as it helps to clean the data before the application of the machine learning algorithm. In this study, we performed several preprocessing steps to address common data issues:

1. **Handling Missing Values:** In medical datasets, it is not uncommon to find that some data are either incomplete or not available at all. In this case, imputation techniques are used to deal with the missing values. In the case of numerical observations, we apply mean imputation to continuous values like age and GFR with missing values. This approach assists in maintaining an appropriate general balance in the distribution of data while preventing significant biases. For the categorical variables like gender and FUNC_STAT_TCR the mode imputation is applied which replaces the missing values by the most frequently occurring values.

2. **Outlier Detection:** Sometimes, extreme values that are highly influential to the dependent variable of the machine learning models particularly in a range measure such as GFR and serum creatinine. Outliers are handled by using statistical methods including the Z-score and IQR whereby the values that fall beyond a given range are excluded from the dataset.

3. *Data Scaling:* This is done because many machine learning algorithms such as Support Vector Machines (SVM) and logistic regression are sensitive to the scale of input features. This process standardizes the features to have zero mean and unit variance prevents any single feature from dominating due to its large magnitude. This is especially so when the features have different units or orders, such as age and

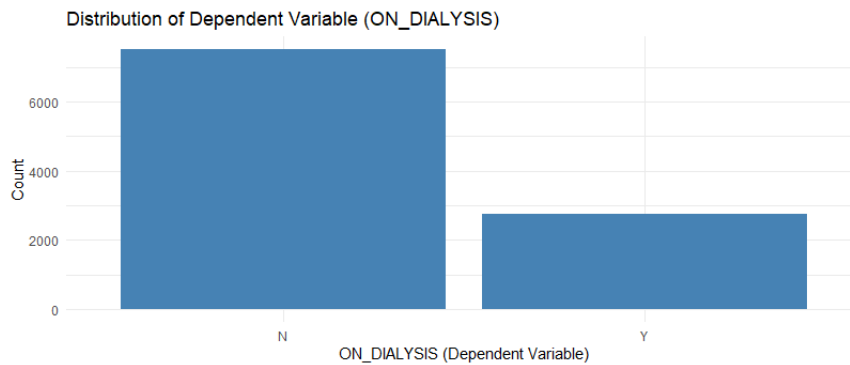
5.0 Results

5.0 Logistic Regression Model: Prediction of Dialysis Dependency.

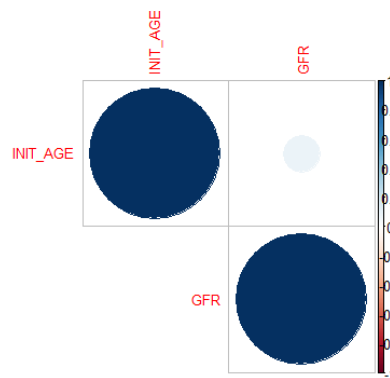
We performed a Logistic Regression model to predict if a patient would need dialysis (ON_DIALYSIS) based on clinical features like age (INIT_AGE), functional status (FUNC_STAT_TCR), gender (GENDER), and glomerular filtration rate (GFR). Some important clinical factors included age, functional status, gender, and GFR, as well as the target variable ON_DIALYSIS.

Exploratory Data Analysis (EDA)

Our EDA began by examining summary statistics of the selected variables. The INIT_AGE variable fell within the range of 18 to 86 years, with a mean age of 54.1 years. The variable FUNC_STAT_TCR had categorical values representing the degree of functionality, from 2080 to 2100. The GFR values represented kidney functions, with values ranging from 0 and 20. Moreover, the target variable ON_DIALYSIS was considered highly unbalanced since only 7,513 patients did not require dialysis, whereas 2,742 did.



We also generated a correlation heatmap to explore the relationship between age and GFR. Although the correlation was weak, there was a slight inverse relationship, suggesting that older patients tend to have lower GFR values, though the association was not strong enough to be definitive.



Logistic Regression Model Training

The logistic regression model was developed to forecast ON_DIALYSIS employing INIT_AGE, FUNC_STAT_TCR, GENDER, and GFR as independent variables. The model was articulated as follows:

$$\text{ON_DIALYSIS} \sim \text{INIT_AGE} + \text{FUNC_STAT_TCR} + \text{GENDER} + \text{GFR}$$

The model was trained on 5-fold cross-validation to ensure the generalizability of the model. The results of the logistic regression model give important insights into the relationship between the predictors and the probability of needing dialysis.

- The intercept was highly significant (p-value < 2e-16), showed a significant baseline probability of patients requiring dialysis.
- The coefficient for age (INIT_AGE) was negative -0.00898 which implies that the probability to initiate dialysis decreases as age rises. This relationship was significant, though contrary to conventional clinical understanding that elderly patient is more likely to require dialysis.
- Functional status (FUNC_STAT_TCR) was also offered a different level of significance. For instance, patients of FUNC_STAT_TCR2010 had a probability of 2.298 to have dialysis more than other patients.
- With regards to gender (GENDER), males had larger odds of requiring dialysis than females (coefficient = 0.475, p-value < 2e-16).
- GFR was also significantly and negatively associated with the probability of requiring dialysis (coefficient = -0.318) which is consistent with clinical intuition that lower GFR, which is a sign of kidney compromise, would correspond to higher dialysis need.

2 Random Forest Model for Predicting GFR

The dataset was divided into an 80% training subset and a 20% testing subset.

Then, the Random Forest model was trained with the formula:

$$\text{GFR} \sim \text{INIT_AGE} + \text{GENDER} + \text{FUNC_STAT_TCR}.$$

The model was developed on 500 trees, with a single variable used per split in creating the decision trees. The Mean Squared Error of the model was found to be 21.89, which is the average squared difference between the actual and predicted values of Glomerular Filtration Rate (GFR). Also, 2.47% of the variance in GFR was explained by the model; that is, the predictors under the model are relatively weak.

Predictions and Age Group Analysis

Once model training was complete, it was used to predict GFR values on the test data. These predictions were stratified by age group to make the predictions easier to interpret. We divided INIT_AGE into five-year intervals (15-20, 20-25, 25-30, 30-35, 35-40, 45-50, 55-60, etc). The stratified data was then summarized for the average age (Avg_INIT_AGE), the average actual GFR (Avg_GFR), and the average predicted GFR (Avg_Prediction) for each respective group. The results of this summarization are presented in the table below, which shows the mean GFR alongside the model's predictions corresponding to each age group:

Age Group	Avg Age	Avg Actual GFR	Avg Predicted GFR
15-20	18.7	14.3	12.7
20-25	22.4	12.6	12.9
25-30	27.4	13.0	13.0
30-35	32.1	12.8	13.0
35-40	37.1	12.6	13.1
40-45	42.1	13.7	13.2
45-50	47.1	12.8	13.2
50-55	52.1	13.3	13.3
55-60	57.0	13.1	13.3
60-65	62.0	13.9	13.5
65-70	67.0	13.7	13.6

70-75	71.8	14.2	13.7
75-80	76.4	12.8	14.0
NA	82.0	16.0	13.9

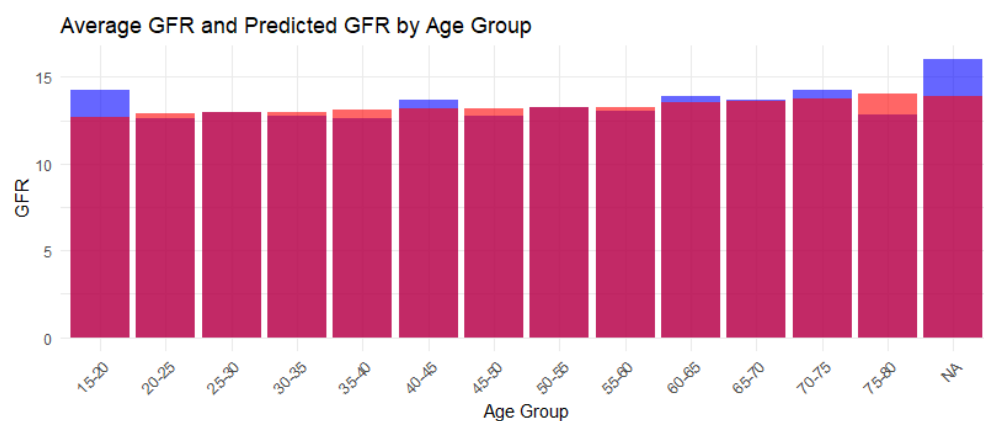
The table shows that in younger age groups (e.g., 15-20 years), the predicted GFR was generally lower than the actual GFR, while in older age groups, the predictions were closer to the true values.

Visualizations of Model Performance

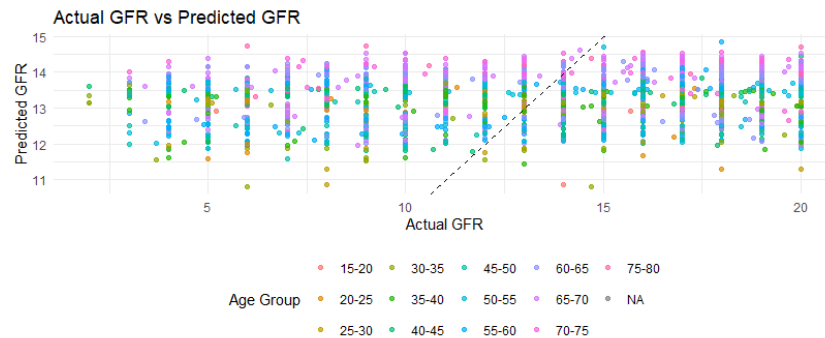
Several plots were created to visualize the performance of the Random Forest model:

1. **Bar Plot Demonstrating Average GFR and Predicted GFR by Age Group:**

The bar plot represents a visual comparison between the average actual and predicted GFR for different age groups. In this graph, blue bars show the average of the actual GFR, while the red ones represent the predicted GFR. The plot shows that predictions were closer to the true values in older age groups, while younger groups exhibited slight underprediction. The x-axis of this plot represents the age groups, and the y-axis represents the values of GFR.



2. **Scatter Plot of Actual vs. Predicted GFR:** The scatter plot showed the actual and predicted GFR values; different color was used for each age group. The dashed diagonal lines represented the amount of correct prediction where the slope is equal to 1. Most of the points cluster near the 1:1 line which means the model achieved good prediction, but there are few noticeable cases of under-prediction of GFR.



3. Logistic Regression Model for Donation Prediction

The logistic regression model was built to predict the likelihood of donation based on gender, age group, and education level using the formula:

$$\text{DONATION} \sim \text{GENDER} + \text{AGE_GROUP} + \text{EDUCATION}$$

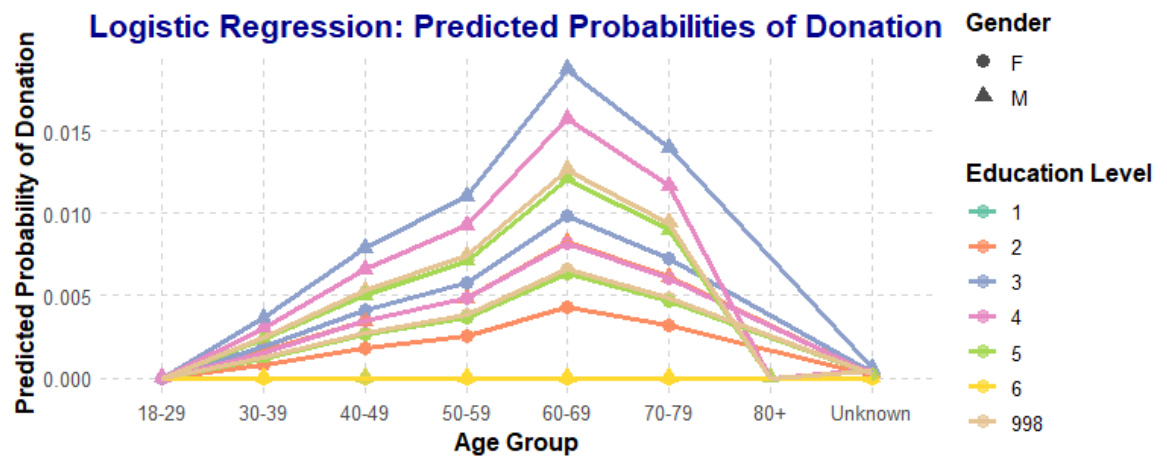
Model Summary

The summary of the logistic regression model showed the following coefficients and significance levels:

- Intercept had a very large negative value (-38.92) and hence very high standard errors which showed that there was a problem with model fit because of the rare outcome or due to imbalance data.
- The coefficient for **gender** (0.656) was not significant, with a p-value of **0.132**.

- The age group and education variables also exhibited high standard errors and p-values, indicating that these variables did not significantly affect the probability of donation in this model.

Visualization of Results



To interpret the logistic regression findings more conveniently we created a scatter plot of the predicted probability of donating with the AGE_GROUP variable and the EDUCATION and GENDER as aesthetics. The plot depicted that education and gender moderated the effect of the age group on the predicted probability of donation.

The plot showed that in the vast majority of age subgroups, the predicted probability of the donation was nearly the same for all levels of education and gender.

References

- Bellini, M. I., Nozdrin, M., Pengel, L., Knight, S., & Papalois, V. (2021). The impact of recipient demographics on outcomes from living donor kidneys: systematic review and meta-analysis. *Journal of clinical medicine*, 10(23), 5556.
- Burlacu, A., Iftene, A., Jugrin, D., Popa, I. V., Lupu, P. M., Vlad, C., & Covic, A. (2020). Using artificial intelligence resources in dialysis and kidney transplant patients: a literature review. *BioMed research international*, 2020(1), 9867872.
- Chen, Q., Zhang, Y., Zhang, M., Li, Z., & Liu, J. (2022). Application of machine learning algorithms to predict acute kidney injury in elderly orthopedic postoperative patients. *Clinical Interventions in Aging*, 317-330.
- Gogoi, P., & Valan, J. A. (2024). Machine learning approaches for predicting and diagnosing chronic kidney disease: current trends, challenges, solutions, and future directions. *International Urology and Nephrology*, 1-24.
- Gotlieb, N., Azhie, A., Sharma, D., Spann, A., Suo, N. J., Tran, J., ... & Bhat, M. (2022). The promise of machine learning applications in solid organ transplantation. *NPJ digital medicine*, 5(1), 89.
- Kherabi, Y., Thy, M., Bouzid, D., Antcliffe, D. B., Rawson, T. M., & Peiffer-Smadja, N. (2024). Machine learning to predict antimicrobial resistance: future applications in clinical practice?. *Infectious Diseases Now*, 104864.
- Mulugeta, G., Zewotir, T., Tegegne, A. S., Juhar, L. H., & Muleta, M. B. (2024). Developing Clinical Prognostic Models to Predict Graft Survival after Renal Transplantation: Comparison of Statistical and Machine Learning Models.

- Pakaya, R. E., Syam, Y., & Syahrul, S. (2021). Analysis of self-efficacy and characteristics of patients with chronic kidney disease underwent hemodialysis. *Jurnal Keperawatan*, 12(1), 93-98.
- Pan, Y. J. (2023). Building a Predictive Model for Kidney Replacement Therapy Occurrence Using Machine Learning and Transfer Learning (Master's thesis, National Yang Ming Chiao Tung University).
- Sharif, A. (2022). Deceased donor characteristics and kidney transplant outcomes. *Transplant International*, 35, 10482.
- Youmans, Q., Aimo, A., & Emdin, M. (2023). Race-Independent Estimates of Glomerular Filtration Rate and Heart Failure Risk: A Step Toward Equity in Heart Failure?. *Journal of Cardiac Failure*.